Case Study

Institution: Shri Sant Gajanan Maharaj College of Engineering, Shegaon
Department: Electrical (Electronics & Power) Engineering
Course Title & Code: Electronic Devices and Circuits (3EP05)
Class: Second Year Electrical Engineering
Semester: III
Course Teacher: Mr. G. N. Bonde
Title of the Innovative Practice: Virtual Lab for Electronic Devices and Circuits

Acivity Link: https://be-iitkgp.vlabs.ac.in/List%20of%20experiments.html

Objectives / Goals of the Practice

1. **Enhance Conceptual Understanding of EDC Fundamentals**
   - To provide interactive, simulation-based experiments for better understanding of diode and transistor characteristics.
2. **Bridge the Gap Between Theory and Practical Application**
   - Enable students to visualize real-time device behavior under varying input conditions through virtual simulations.
3. **Foster Independent Learning and Technical Confidence**
   - Encourage students to perform circuit simulations independently, improving their confidence in analyzing and designing electronic circuits.

Methods and Strategies Employed

- **Adoption of Virtual Simulation Tools:**
  Software platforms like NI Multisim, Tinkercad, or online circuit simulators were used for executing basic EDC experiments.
- **Hands-on Practice with Circuit Building and Analysis:**
  Students virtually constructed circuits, observed VI characteristics, and analyzed performance under different biasing conditions.
- **Analytical Assignments & Error Correction Tasks:**
  Students interpreted output waveforms, plotted characteristics, and identified/corrected errors in circuit setups.

List of Experiments Performed

1. **VI Characteristics of a Diode**
   - Study the forward and reverse bias characteristics using simulation.
2. **Half Wave Rectification**
   - Analyze the output waveform and efficiency of a half-wave rectifier circuit.
3. **Full Wave Rectification**
   - Compare center-tap and bridge-type rectifiers using simulation tools.
4. **Zener Diode as Voltage Regulator**
   - Observe voltage regulation performance across varying load and input conditions.

5. **BJT Common Emitter Characteristics**
    - Plot input and output characteristics and determine current gain (β).
6. **BJT Common Base Characteristics**
    - Analyze input and output characteristics and compare with CE configuration.

Effective Presentation Strategy

1. **Foundational Teaching:**
    - Delivered core theoretical concepts during regular lectures.
2. **Interactive Demonstrations:**
    - Live simulation demos were conducted by the faculty to show circuit behavior.
3. **Hands-on Virtual Lab Practice:**
    - Students practiced and verified circuit behavior using provided simulation tools.
4. **Collaborative Review and Evaluation:**
    - Group discussions and Q&A sessions ensured deeper understanding and peer learning.

Program Outcomes (POs) & Program Specific Outcomes (PSOs) Mapped

- **POs:** PO1, PO2, PO3, PO4, PO5, PO12
- **PSOs:** PSO1, PSO2

Reflective Critique

- **Initial Hurdles:**
  Students were unfamiliar with simulation tools, but became proficient with guided practice.
- **Improved Conceptual Clarity:**
  Visualization of waveforms and characteristic curves helped students internalize theoretical principles.
- **Suggestions from Faculty:**
  Recommended periodic integration of simulation-based assignments in regular lab curriculum for consistency.

Evidence of Success

- Over **65% of students** successfully completed and submitted all virtual experiments.
- Improved comprehension observed during oral assessments and in written lab reports.
- Students expressed increased confidence in analyzing diode and transistor circuits.

Challenges Faced

- **Tool Familiarity:**
  Some students required extra sessions to become comfortable with simulation environments.

- **Assessment Limitation:**
  Monitoring real-time engagement and individual troubleshooting skills was limited in virtual mode.

Figure : Photo of the activity

Feedback Link: https://forms.office.com/r/b55hTRx02x?origin=lprLink